Fiscal Year:	FY 2017	**Task Last Updated:**	FY 07/31/2018
PI Name:	Koochesfahani, Manoochehr Ph.D.		
Project Title:	ZBOT Thermometry and Velocimetry Diagnostics Development		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	E. Lansing	**State:**	MI
Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	02/05/2016	**End Date:**	12/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2018 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 (Ed., 7/31/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pouya, Shahram Ph.D. (Michigan State University)		
Grant/Contract No.:	NNX16AD52A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In-orbit or on-surface cryogenic storage for propellant and life support consumables is essential to all stages of NASA's Exploration Program. However, heat leakage from the surroundings results in liquid vaporization and significant mass loss and/or significant increase in the storage tank pressure. To mitigate the unwanted effects of cryogen vaporization, the Zero Boil-Off (ZBO) concept is being considered as an innovative approach for ventless means of controlling tank pressure through the application of active heat removal and forced liquid jet mixing. The Zero Boil-Off Tank (ZBOT) experiment is designed for operation on the International Space Station (ISS). Its purpose is to examine the ZBO concept and, along with computational modeling efforts, provide information on the fundamental physics of the underlying complex thermofluids environment, which is needed for engineering design optimization of the ZBO concept. The ZBOT experiment currently relies on Digital Particle Image Velocimetry (DPIV) to obtain flow field information and visualization in the test fluid, Perfluoro-n-Pentane (PnP). Difficulties have been encountered, however, in using DPIV. These difficulties include a drastic loss of particles over time and decrease of fluorescence intensity over time from the fluorescent-coated particles, along with concerns over particle sticking and agglomeration. The proposed effort is motivated by the need to obtain non-intrusive, whole-field, temperature and/or velocity data using optical diagnostics in the ZBOT microgravity experiment in order to supplement the computational modeling efforts in progress. Because of the challenges experienced by the ZBOT team in using particle-based techniques (i.e., DPIV), our focus will be on development of molecular-based approaches for measurements within PnP. These include laser induced fluorescence (LIF) methods for thermometry and molecular tagging approaches for velocimetry (MTV).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Ed. note (July 2018): Compiled from Principal Investigator's annual technical report submitted to management in February 2017. The primary focus of the current effort has been on finding and/or synthesizing dyes/tracers that can be solubilized in PnP and have properties useful for thermometry or velocimetry techniques. As mentioned in the proposal, we had identified Coumarin 153 as a fluorescent dye with potential for thermometry in ZBOT experiments. We have now quantified its various photochemical properties and have concluded that its susceptibility to photobleaching combined with its relatively weak thermal sensitivity renders it an undesirable tracer. However, its bleaching characteristics make Coumarin a suitable dye for velocimetry with MTV. We have conducted proof-of-concept experiments showing viability of MTV measurement using this approach. In parallel, we have continued our effort to synthesize compounds with fluorinated functionality to dissolve in PnP. To this end, the Blanchard Lab has successfully modified nanocrystal quantum dots (QD) to dissolve into PnP and our preliminary measurements show promising temperature sensitivity to be used as the main thermometry tracer. Calibration experiments have shown a two-color LIF thermometry approach as a ratiometric technique is feasible.
Bibliography Type:	Description: (Last Updated: 03/02/2026)
Abstracts for Journals and Proceedings	Pouya S, Blanchard G, Koochesfahani M. "Development of molecular based optical techniques for thermometry and velocimetry for fluorocarbon media." 69th Annual Meeting of the APS Division of Fluid Dynamics, Portland, OR, November 20–22, 2016. Bulletin of the American Physical Society. 2016 Nov;61(20):Abstract: R31.00004. http://meetings.aps.org/link/BAPS.2016.DFD.R31.4 , Nov-2016